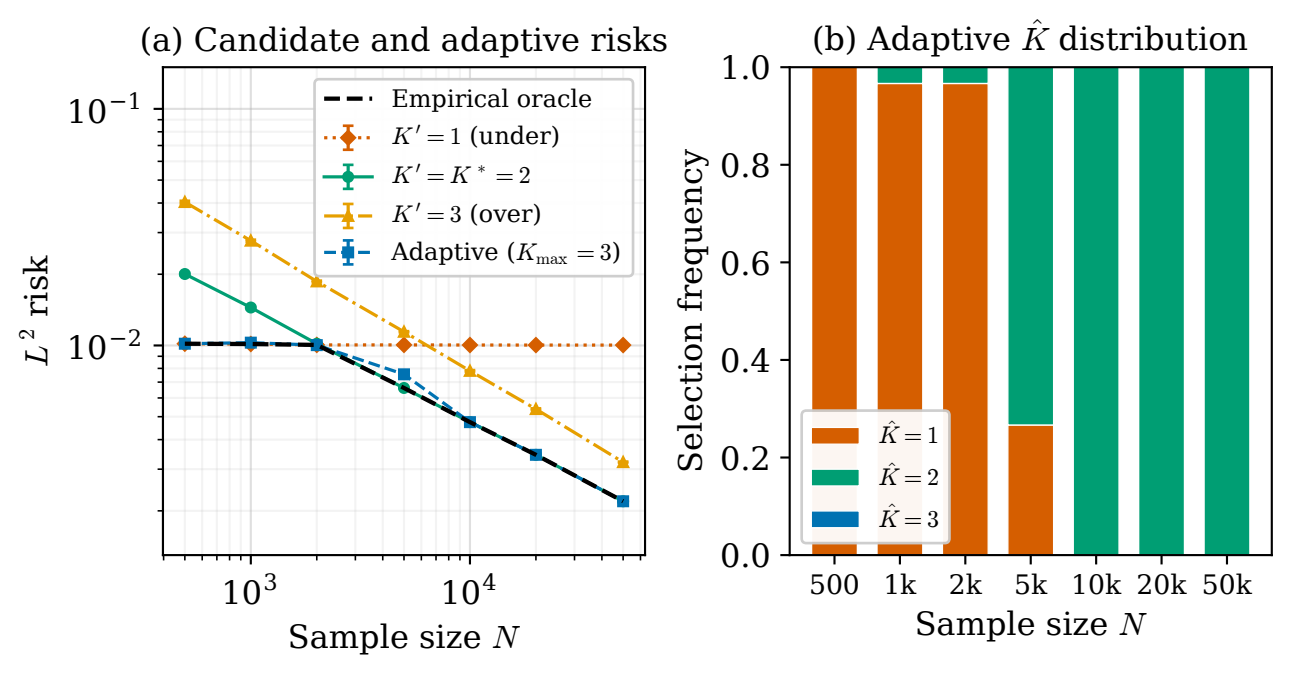
(a) Candidate and adaptive risks
Empirical oracle
$K' = 1$ (under)
$K' = K^* = 2$
$K' = 3$ (over)
Adaptive ($K_{\max} = 3$)
L^2 risk
10^{-1}
10^{-2}
10^3
10^4
Sample size N
(b) Adaptive $\hat{K}$ distribution
Selection frequency
1.0
0.8
0.6
0.4
0.2
0.0
$\hat{K} = 1$
$\hat{K} = 2$
$\hat{K} = 3$
500
1k
2k
5k
10k
20k
50k
Sample size N